

PRACTICAL NO 1**AIM:-**

1A.Design and implement algorithms to encrypt and decrypt messages using classical substitution techniques

```
def encrypt(text, shift):
    result = ""
    for char in text:
        if char.isalpha():
            start = ord('A') if char.isupper() else ord('a')
            result += chr((ord(char) - start + shift) % 26 + start)
        else:
            result += char
    return result

def decrypt(text, shift):
    return encrypt(text, -shift)

print("=== Caesar Cipher ===")
print("1. Encrypt")
print("2. Decrypt")

choice = input("Enter Choice: ")

message = input("Enter Message: ")
shift = int(input("Enter Shift Value: "))

if choice == "1":
    print("Encrypted Message:", encrypt(message, shift))
elif choice == "2":
    print("Decrypted Message:", decrypt(message, shift))
else:
    print("Invalid Choice")
```

OUTPUT:-

```

Python 3.14.3 (tags/v3.14.3:323c59a, Feb 3 2026, 16:04:56) [MSC v.1944 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>>
===== RESTART: C:\Users\as993\Downloads\PRACTICAL NO 1\Caesar_CLI.py =====
=== Caesar Cipher ===
1. Encrypt
2. Decrypt
Enter Choice: 1
Enter Message: ABHISHEK DINESH SINGH
Enter Shift Value: 3
Encrypted Message: DEKLVKHN GLQHVK VLQJK
>>>
===== RESTART: C:\Users\as993\Downloads\PRACTICAL NO 1\Caesar_CLI.py =====
=== Caesar Cipher ===
1. Encrypt
2. Decrypt
Enter Choice: 2
Enter Message: DEKLVKHN GLQHVK VLQJK
Enter Shift Value: 3
Decrypted Message: ABHISHEK DINESH SINGH
>>>

```

CODE:-

```

import tkinter as tk

def encrypt(text, shift):
    result = ""
    for char in text:
        if char.isalpha():
            start = ord('A') if char.isupper() else ord('a')
            result += chr((ord(char) - start + shift) % 26 + start)
        else:
            result += char
    return result

def decrypt():
    result.set(encrypt(message.get(), -int(shift.get())))

def encrypt_message():
    result.set(encrypt(message.get(), int(shift.get())))

root = tk.Tk()
root.title("Caesar Cipher")

tk.Label(root, text="Message").pack()
message = tk.Entry(root, width=40)

```

ABHISHEK DINESH SINGH

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```
message.pack()

tk.Label(root, text="Shift").pack()
shift = tk.Entry(root)
shift.pack()

result = tk.StringVar()

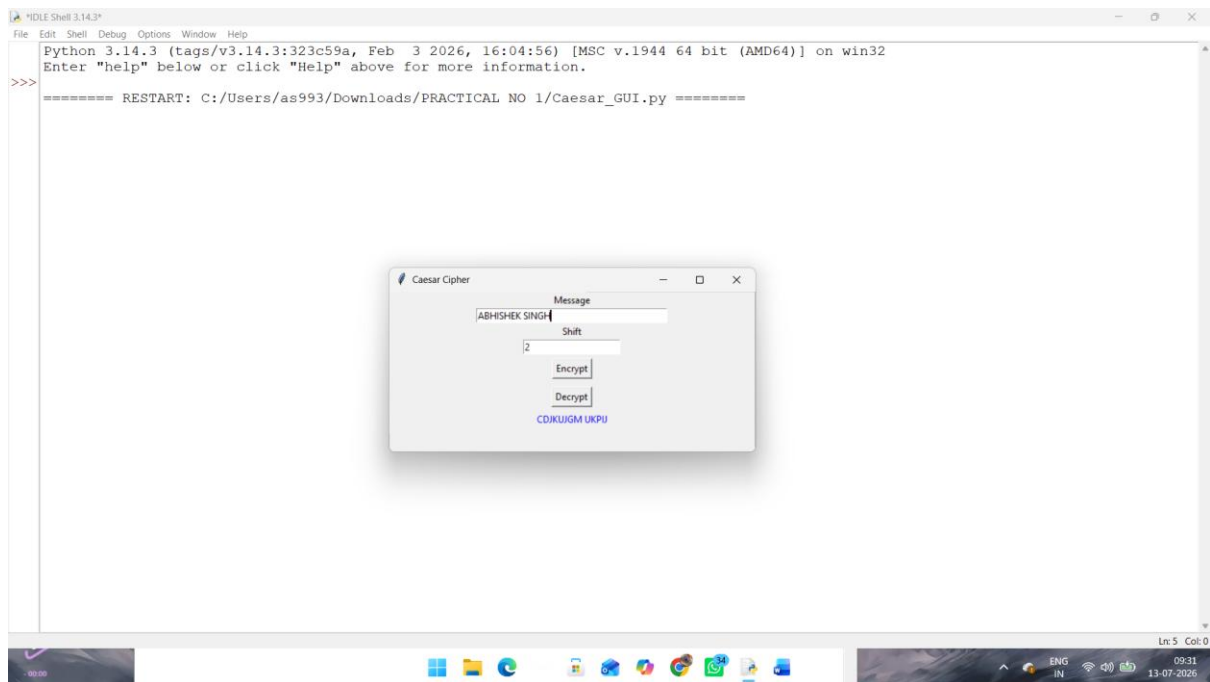
tk.Button(root, text="Encrypt", command=encrypt_message).pack(pady=5)
tk.Button(root, text="Decrypt", command=decrypt).pack(pady=5)

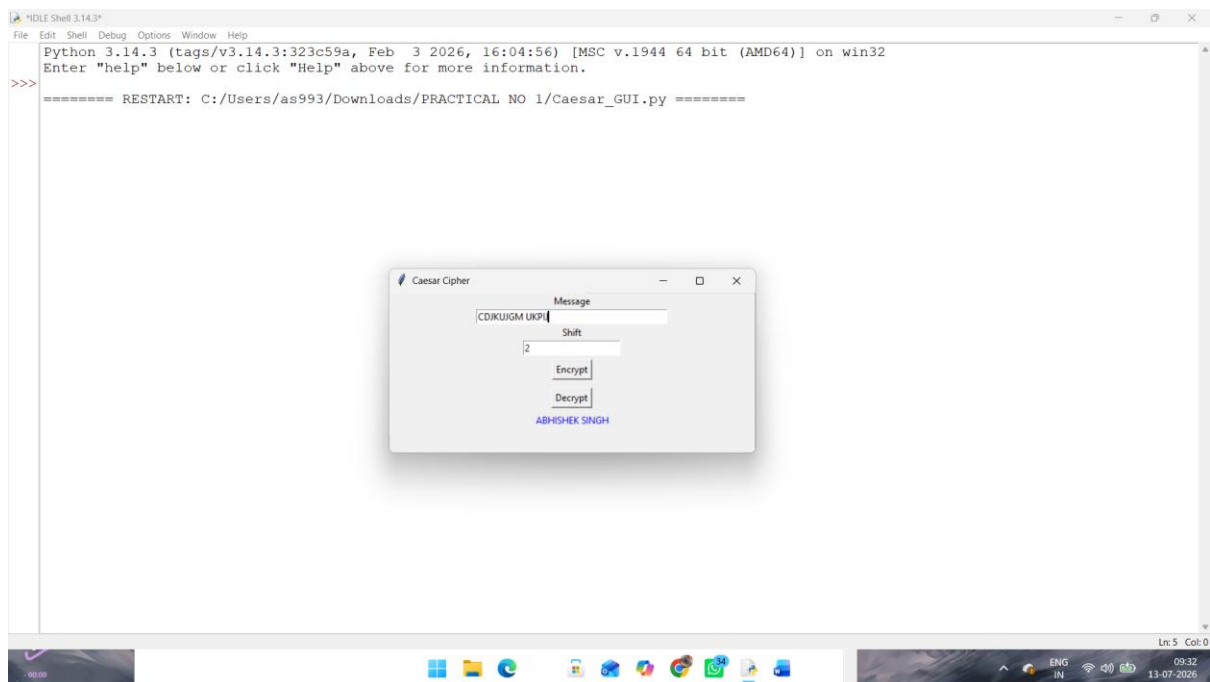
tk.Label(root, textvariable=result, fg="blue").pack()

root.mainloop()
```

OUTPUT:

ENCRYPT



DECRYPT**1B.Design and implement algorithms to encrypt and decrypt messages using transposition techniques****CODE:-**

```
def encrypt(text):
    return text[::-1]

def decrypt(text):
    return text[::-1]

print("=== Transposition Cipher ===")
print("1. Encrypt")
print("2. Decrypt")

choice = input("Enter Choice: ")

message = input("Enter Message: ")

if choice == "1":
    print("Encrypted Message:", encrypt(message))
elif choice == "2":
    print("Decrypted Message:", decrypt(message))
else:
    print("Invalid Choice")
```

OUTPUT:

```

Python 3.14.3 (tags/v3.14.3:323c59a, Feb 3 2026, 16:04:56) [MSC v.1944 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>>
===== RESTART: C:/Users/as993/Downloads/PRACTICAL NO 1/Transposition_CLI.py =====
==== Transposition Cipher ====
1. Encrypt
2. Decrypt
Enter Choice: 1
Enter Message: ABHISHEK DINESH SINGH
Encrypted Message: HGNIS HSENIID KEHSIHBA
>>>
===== RESTART: C:/Users/as993/Downloads/PRACTICAL NO 1/Transposition_CLI.py =====
==== Transposition Cipher ====
1. Encrypt
2. Decrypt
Enter Choice: 2
Enter Message: HGNIS HSENIID KEHSIHBA
Decrypted Message: ABHISHEK DINESH SINGH
>>>

```

CODE:-

```

import tkinter as tk

def encrypt():
    result.set(message.get()[::-1])

def decrypt():
    result.set(message.get()[::-1])

root = tk.Tk()
root.title("Transposition Cipher")

tk.Label(root, text="Message").pack()

message = tk.Entry(root, width=40)
message.pack()

result = tk.StringVar()

tk.Button(root, text="Encrypt", command=encrypt).pack(pady=5)
tk.Button(root, text="Decrypt", command=decrypt).pack(pady=5)

tk.Label(root, textvariable=result, fg="blue").pack()

root.mainloop()

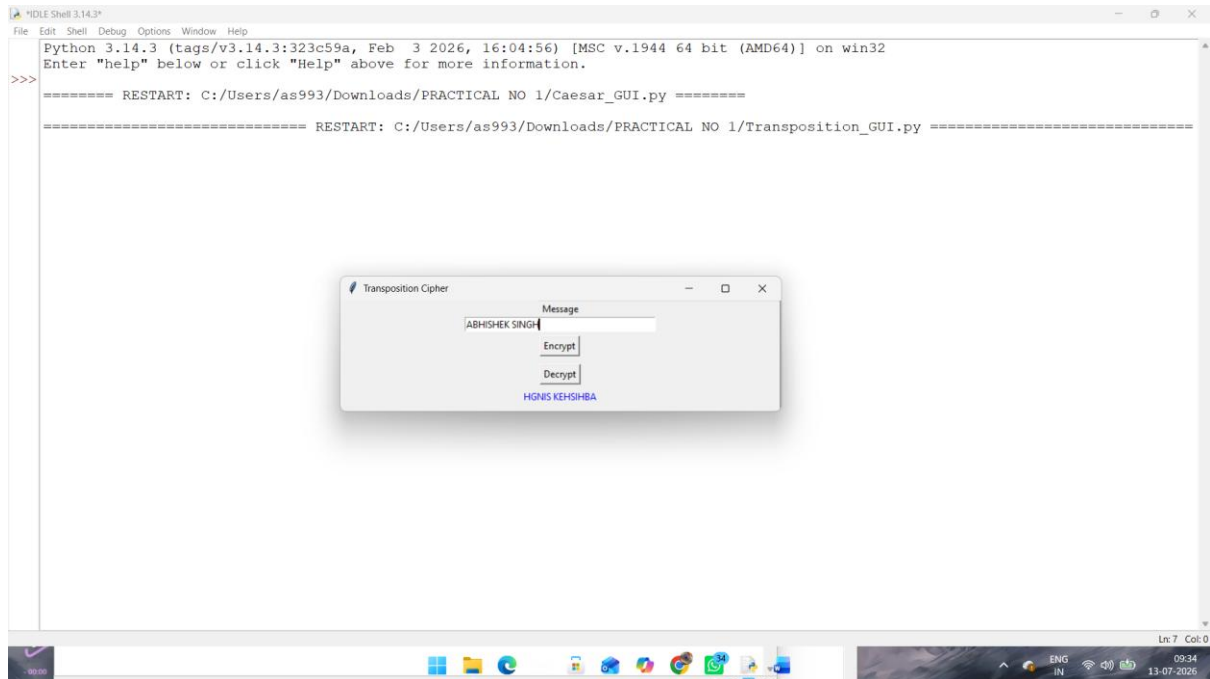
```

ABHISHEK DINESH SINGH

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OUTPUT

ENCRYPT



DECRYPT

